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United States Patent Application Publication

20250284409

Kind Code

A1

Publication Date

September 11, 2025

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TWO-STAGE EMERGENCY DATA STORING OPERATION

Abstract

Implementations described herein relate to a two-stage emergency data storing operation. In some implementations, a memory device may detect a power loss notification signal that indicates a power loss condition of the memory device. The memory device may read a mode register bit of the memory device that indicates to perform a data storing operation that includes a first data storing stage and a second data storing stage. The first data storing stage may include storing data associated with the memory device prior to the memory device experiencing a power loss, and the second data storing stage may include storing data and metadata associated with the memory device prior to the memory device experiencing the power loss. The memory device may initiate the data storing operation and may selectively acknowledge the power loss condition based on completing the first data storing stage or the second data storing stage.

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Family ID: 1000008621195

Appl. No.: 19/217243

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18511446 20231116 parent-grant-document US 12314573 child US 19217243

us-provisional-application US 63385434 20221130

Publication Classification

Int. Cl.: G06F3/06 (20060101)

U.S. Cl.:

CPC **G06F3/0619** (20130101); **G06F3/0634** (20130101); **G06F3/0659** (20130101);
G06F3/0673 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/511,446, filed Nov. 16, 2023 (now U.S. Pat. No. 12,314,573), which claims priority to U.S. Provisional Patent Application No. 63/385,434, filed on Nov. 30, 2022, entitled “TWO-STAGE EMERGENCY DATA STORING OPERATION,” and assigned to the assignee hereof. The contents of the above-referenced applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure generally relates to memory devices, memory device operations, and, for example, to a two-stage emergency data storing operation.

BACKGROUND

[0003] Memory devices are widely used to store information in various electronic devices. A memory device includes memory cells. A memory cell is an electronic circuit capable of being programmed to a data state of two or more data states. For example, a memory cell may be programmed to a data state that represents a single binary value, often denoted by a binary “1” or a binary “0.” As another example, a memory cell may be programmed to a data state that represents a fractional value (e.g., 0.5, 1.5, or the like). To store information, an electronic device may write to, or program, a set of memory cells. To access the stored information, the electronic device may read, or sense, the stored state from the set of memory cells.

[0004] Various types of memory devices exist, including random access memory (RAM), read only memory (ROM), dynamic RAM (DRAM), static RAM (SRAM), synchronous dynamic RAM (SDRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), holographic RAM (HRAM), flash memory (e.g., NAND memory and NOR memory), and others. A memory device may be volatile or non-volatile. Non-volatile memory (e.g., flash memory) can store data for extended periods of time even in the absence of an external power source. Volatile memory (e.g., DRAM) may lose stored data over time unless the volatile memory is refreshed by a power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram illustrating an example system capable of emergency data storing operation selection.

[0006] FIG. 2 is a diagram of example components included in a memory device.

[0007] FIG. 3 is a diagram illustrating an example of emergency data storing.

[0008] FIGS. 4A and 4B illustrate examples of a two-stage emergency storing operation.

[0009] FIG. 5 is a diagrammatic view of an example memory device.

[0010] FIG. 6 is a flowchart of an example method associated with emergency data storing operation selection.

DETAILED DESCRIPTION

[0011] A memory device may experience a low battery condition and/or a power loss condition. The low battery condition of the memory device may occur when a battery that is providing power

to the memory device is in a low power state. For example, a cellular phone may include a memory device and a battery that provides power to the memory device. The memory device may experience a low battery condition when a charge of the battery is less than a threshold, such as less than ten percent, less than five percent, or less than two percent, among other examples. The power loss condition of the memory device may occur when the charge of the battery is zero percent or close to zero percent. For example, a memory device that is experiencing a low battery condition may experience a power loss condition if the battery that is powering the memory device (and/or other components) continues to be used. Alternatively, the power loss condition may occur when the battery becomes disconnected from the memory device. For example, when a vehicle that includes a battery and a memory device crashes, the battery may become disconnected from the memory device, and the memory device may no longer receive power from the battery.

[0012] A memory device experiencing the low battery condition or the power loss condition may perform an emergency storing operation to prevent data loss. In some cases, the emergency storing operation may be an emergency power fail (EPF) operation. In this case, the memory device may store data (such as in-flight data) prior to a power loss acknowledgement (PLA) signal being asserted, but may not store metadata (such as logical block address (LBA) information) prior to the PLA being asserted. This may require a longer boot-up time during a next powering on of the memory device, for example, to enable the memory device to recover the metadata. In some cases, the emergency storing operation may be a forced quiescence (FQ) operation. In this case, the memory device may store both data and metadata prior to the PLA being asserted, and the memory device may not need the longer boot-up time at the next powering on of the memory device. The memory device may use power from one or more capacitors associated with the memory device to perform the EPF operation or the FQ operation. In some cases, a register setting may indicate which emergency storing operation is to be used. For example, a first bit of the register may indicate for the memory device to use the EPF operation to store the data prior to a memory device shut-down, and a second bit of the register may indicate for the memory device to use the FQ operation to store the data and the associated metadata prior to the memory device shut-down.

[0013] In some cases, it may not be desirable for the memory device to use the EPF operation due to the longer boot-up times associated with the EPF operation, particularly when there is enough time for an FQ operation to be performed. For example, a user of the automobile may not be able to wait a time period for the memory device to recover the metadata upon a next powering on of the memory device prior to being able to drive the automobile. In some cases, protocol level commands may be used to control or manage the emergency storing operations. For example, the system may include a power management integrated circuit (PMIC) triggering a power loss management interrupt routine, a host system-on-chip (SOC) executing an interrupt service routine, and a memory controller serving a non-volatile memory express (NVMe) command associated with the emergency storing operation into a command queue. This may result in additional delays for the memory device with respect to storing the data and/or the metadata. In some cases, a design architecture of the PMIC may result in the host SOC being powered off during an APL to maximize a power hold-off time for preventing data corruption. However, there is currently no method that enables the memory device to control (e.g., using hardware signals) which emergency storing operation is to be performed based on a low battery condition or a power loss condition, such as based on detection of a power loss notification (PLN) signal.

[0014] In some implementations, the memory device may perform (or attempt to perform) both the EPF operation and the FQ operation during an emergency storing operation. The memory device may detect a PLN signal that indicates a power loss condition or a low battery condition of the memory device. The memory device may read a mode register bit of the memory device that indicates to perform a data storing operation that includes the EPF operation and the FQ operation. The memory device may initiate the data storing operation that includes the EPF operation and the FQ operation based on detecting the PLN signal and based on the register bit indicating to perform

the data storing operation that includes the EPF operation and the FQ operation. For example, the memory device may initiate the EPF operation to store data associated with the memory device prior to the memory device experiencing a power loss. If the memory device completes the EPF operation, the memory device may proceed to perform the FQ operation. As a result, the memory device may perform the EPF operation to save data prior to the memory device experiencing the power loss, and may continue to perform the FQ operation upon completion of the EPF operation. This may enable the memory device to store data and metadata prior to the memory device experiencing a power loss, thereby preventing a longer a boot-up time during a next powering on of the memory device. Additional details are described herein.

[0015] FIG. 1 is a diagram illustrating an example system **100** capable of emergency data storing operation selection. The system **100** may include one or more devices, apparatuses, and/or components for performing operations described herein. For example, the system **100** may include a host device **110** and a memory device **120**. The memory device **120** may include a controller **130** and memory **140**. The host device **110** may communicate with the memory device **120** (e.g., the controller **130** of the memory device **120**) via a host interface **150**. The controller **130** and the memory **140** may communicate via a memory interface **160**.

[0016] The system **100** may be any electronic device configured to store data in memory. For example, the system **100** may be a computer, a mobile phone, a wired or wireless communication device, a network device, a server, a device in a data center, a device in a cloud computing environment, a vehicle (e.g., an automobile or an airplane), and/or an Internet of Things (IoT) device. The host device **110** may include one or more processors configured to execute instructions and store data in the memory **140**. For example, the host device **110** may include a central processing unit (CPU), a graphics processing unit (GPU), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), and/or another type of processing component.

[0017] The memory device **120** may be any electronic device configured to store data in memory. In some implementations, the memory device **120** may be an electronic device configured to store data temporarily in volatile memory. For example, the memory device **120** may be a random-access memory (RAM) device, such as a dynamic RAM (DRAM) device or a static RAM (SRAM) device. In this case, the memory **140** may include volatile memory that requires power to maintain stored data and that loses stored data after the memory device **120** is powered off. For example, the memory **140** may include one or more latches and/or RAM, such as DRAM and/or SRAM. In some implementations, the memory **140** may include non-volatile memory configured to maintain stored data after the memory device **120** is powered off, such as NAND memory or NOR memory. For example, the non-volatile memory may store persistent firmware or other instructions for execution by the controller **130**.

[0018] The controller **130** may be any device configured to communicate with the host device (e.g., via the host interface **150**) and the memory **140** (e.g., via the memory interface **160**). Additionally, or alternatively, the controller **130** may be configured to control operations of the memory device **120** and/or the memory **140**. For example, the controller **130** may include a memory controller, a system controller, an ASIC, an FPGA, a processor, a microcontroller, and/or one or more processing components. In some implementations, the controller **130** may be a high-level controller, which may communicate directly with the host device **110** and may instruct one or more low-level controllers regarding memory operations to be performed in connection with the memory **140**. In some implementations, the controller **130** may be a low-level controller, which may receive instructions regarding memory operations from a high-level controller that interfaces directly with the host device **110**. As an example, a high-level controller may be an SSD controller, and a low-level controller may be a non-volatile memory controller (e.g., a NAND controller) or a volatile memory controller (e.g., a DRAM controller). In some implementations, a set of operations described herein as being performed by the controller **130** may be performed by a single controller (e.g., the entire set of operations may be performed by a single high-level controller or a single

low-level controller). Alternatively, a set of operations described herein as being performed by the controller **130** may be performed by more than one controller (e.g., a first subset of the operations may be performed by a high-level controller and a second subset of the operations may be performed by a low-level controller).

[0019] The host interface **150** enables communication between the host device **110** and the memory device **120**. The host interface **150** may include, for example, a Small Computer System Interface (SCSI), a Serial-Attached SCSI (SAS), a Serial Advanced Technology Attachment (SATA) interface, a Peripheral Component Interconnect Express (PCIe) interface, an NVMe interface, a USB interface, a Universal Flash Storage (UFS) interface, and/or an embedded multimedia card (eMMC) interface.

[0020] The memory interface **160** enables communication between the memory device **120** and the memory **140**. The memory interface **160** may include a non-volatile memory interface (e.g., for communicating with non-volatile memory), such as a NAND interface or a NOR interface.

Additionally, or alternatively, the memory interface **160** may include a volatile memory interface (e.g., for communicating with volatile memory), such as a double data rate (DDR) interface.

[0021] In some implementations, the memory device **120** and/or the controller **130** may be configured to detect a power loss notification signal that indicates a power loss condition of the memory device; read a mode register bit of the memory device that indicates to perform a data storing operation that includes a first data storing stage and a second data storing stage, wherein the first data storing stage includes storing data associated with the memory device prior to the memory device experiencing a power loss, and the second data storing stage includes storing data and metadata associated with the memory device prior to the memory device experiencing the power loss; initiate the data storing operation; and selectively acknowledge the power loss condition based on completing the first data storing stage or the second data storing stage.

[0022] In some implementations, the memory device **120** and/or the controller **130** may be configured to a power management component configured to: detect a power loss condition associated with a memory device; and transmit a power loss notification signal to the memory device that indicates the power loss condition; and the memory device, wherein the memory device is configured to: receive the power loss notification signal; read a mode register bit of the memory device that indicates to perform a data storing operation that includes a first data storing stage and a second data storing stage, wherein the first data storing stage includes storing data associated with the memory device prior to the memory device experiencing a power loss, and the second data storing stage includes storing data and metadata associated with the memory device prior to the memory device experiencing the power loss; initiate the data storing operation; and selectively acknowledge the power loss condition based on completing the first data storing stage or the second data storing stage.

[0023] In some implementations, the memory device **120** and/or the controller **130** may be configured to detect a power loss condition of the memory device; read a mode register bit that indicates to perform a data storing operation that includes an emergency power fail operation and a forced quiescence operation; initiate and completing the emergency power fail operation; initiate the forced quiescence operation based on completing the emergency power fail operation; and assert a power loss acknowledgement based on completing the forced quiescence operation.

[0024] As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

[0025] FIG. **2** is a diagram of example **200** components included in a memory device **120**. As described above in connection with FIG. **1**, the memory device **120** may include a controller **130** and memory **140**. As shown in FIG. **2**, the memory **140** may include one or more non-volatile memory arrays **205**, such as one or more NAND memory arrays and/or one or more NOR memory arrays. Additionally, or alternatively, the memory **140** may include one or more volatile memory arrays **210**, such as one or more SRAM arrays and/or one or more DRAM arrays. The controller

130 may transmit signals to and receive signals from a non-volatile memory array **205** using a non-volatile memory interface **215**. The controller **130** may transmit signals to and receive signals from a volatile memory array **210** using a volatile memory interface **220**.

[0026] The controller **130** may control operations of the memory **140**, such as by executing one or more instructions. For example, the memory device **120** may store one or more instructions in the memory **140** as firmware, and the controller **130** may execute those one or more instructions. Additionally, or alternatively, the controller **130** may receive one or more instructions from the host device **110** via the host interface **150**, and may execute those one or more instructions. In some implementations, a non-transitory computer-readable medium (e.g., volatile memory and/or non-volatile memory) may store a set of instructions (e.g., one or more instructions or code) for execution by the controller **130**. The controller **130** may execute the set of instructions to perform one or more operations or methods described herein. In some implementations, execution of the set of instructions, by the controller **130**, causes the controller **130** and/or the memory device **120** to perform one or more operations or methods described herein. In some implementations, hardwired circuitry is used instead of or in combination with the one or more instructions to perform one or more operations or methods described herein. Additionally, or alternatively, the controller **130** and/or one or more components of the memory device **120** may be configured to perform one or more operations or methods described herein. An instruction is sometimes called a “command.”

[0027] For example, the controller **130** may transmit signals to and/or receive signals from the memory **140** based on the one or more instructions, such as to transfer data to (e.g., write or program), to transfer data from (e.g., read), and/or to erase all or a portion of the memory **140** (e.g., one or more memory cells, pages, sub-blocks, blocks, or planes of the memory **140**). Additionally, or alternatively, the controller **130** may be configured to control access to the memory **140** and/or to provide a translation layer between the host device **110** and the memory **140** (e.g., for mapping logical addresses to physical addresses of a memory array). In some implementations, the controller **130** may translate a host interface command (e.g., a command received from the host device **110**) into a memory interface command (e.g., a command for performing an operation on a memory array).

[0028] As shown in FIG. 2, the controller **130** may include a memory management component **225**, an EPF component **230**, an FQ component **235**, and/or an emergency storing component **240**. In some implementations, one or more of these components are implemented as one or more instructions (e.g., firmware) executed by the controller **130**. Alternatively, one or more of these components may be implemented as dedicated integrated circuits distinct from the controller **130**.

[0029] The memory management component **225** may be configured to manage performance of the memory device **120**. For example, the memory management component **225** may perform wear leveling, bad block management, block retirement, read disturb management, and/or other memory management operations. In some implementations, the memory device **120** may store (e.g., in memory **140**) one or more memory management tables. A memory management table may store information that may be used by or updated by the memory management component **225**, such as information regarding memory block age, memory block erase count, and/or error information associated with a memory partition (e.g., a memory cell, a row of memory, a block of memory, or the like).

[0030] The EPF component **230** may be configured to perform a first emergency storing operation such as an EPF operation. The EPF operation may include storing memory device data such as in-flight memory device data. The EPF operation may not include storing memory device metadata, or may include storing only a portion of the memory device metadata. The EPF operation may include storing the data prior to the memory device **120** experiencing a power loss (e.g., prior to the memory device being powered down). Additionally, or alternatively, the EPF operation may include storing the data prior to the memory device **120** asserting a PLA signal. The EPF operation may result in a longer boot-up time during a next powering on of the memory device **120**. For

example, the memory device **120** may need to store and/or manage the metadata in the memory device **120** prior to the memory device **120** being available for use.

[0031] The FQ component **235** may be configured to perform a second emergency storing operation such as an FQ operation. The FQ operation may include storing memory device data (such as in-flight data) and memory device metadata. The FQ operation may include storing the data and the metadata prior to the memory device experiencing a power loss (e.g., prior to the memory device being powered down). Additionally, or alternatively, the FQ operation may include storing the data and the metadata prior to the memory device **120** asserting a PLA signal. The FQ operation may result in a normal boot-up time the next time the memory device is powered on. For example, the memory device **120** may not need to store and/or manage the metadata in the memory device **120** prior to the memory device being available for use.

[0032] The emergency storing component **240** may be configured to store memory device data (and/or metadata) based on the memory device **120** performing the EPF operation, the FQ operation, or both the EPF operation and the FQ operation. In some implementations, the emergency storing component **240** may determine whether the memory device **120** is to perform the EPF operation, the FQ operation, or both the EPF operation and the FQ operation based on one or more inputs, such as an EPF input, a PLN input, and a PERST input. In some other implementations, the emergency storing component **240** may perform (or attempt to perform) both the EPF operation and the FQ operation.

[0033] One or more devices or components shown in FIG. **2** may be configured to perform operations described elsewhere herein, such as one or more operations and/or methods described in connection with FIGS. **4**, **5A** and **5B**. For example, the controller **130**, the memory management component **225**, the EPF component **230**, the FQ component **235**, and/or the emergency storing component **240** may be configured to perform one or more operations and/or methods for the memory device **120**.

[0034] The number and arrangement of components shown in FIG. **2** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **2**. Furthermore, two or more components shown in FIG. **2** may be implemented within a single component, or a single component shown in FIG. **2** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) shown in FIG. **2** may perform one or more operations described as being performed by another set of components shown in FIG. **2**.

[0035] FIG. **3** is a diagram illustrating an example **300** of emergency data storing. A host SOC **310** may write data to the memory device **120** and/or may read data from the memory device **120**. A power management component **305** may provide power control to the host SOC **310** and/or may provide a supply voltage to the host SOC **310**.

[0036] Additionally, or alternatively, the power management component **305** may provide a supply voltage to the memory device **120**. In some cases, the power management component **305** may assert a PLN signal **315** to indicate a power loss condition associated with the memory device **120**. For example, the power management component **305** may assert the PLN signal **315** based on the memory device **120** experiencing a low battery condition or a disconnected battery condition. In some cases, the memory device **120** may be configured to assert a PLA signal **320**. The PLA signal **320** may indicate to the power management component **305** that the memory device **120** has completed an EPF operation or an FQ operation.

[0037] In some cases, the memory device **120** may include a mode register **325**. The mode register **325** may indicate whether the memory device **120** is to perform the EPF operation or the FQ operation. For example, a first state (such as a low state) may indicate that the memory device **120** is to perform the EPF operation and a second state (such as a high state) may indicate that the memory device **120** is to perform the FQ operation. When the memory device **120** detects the PLN

signal **315**, the memory device **120** may perform an emergency storing operation based on a status of the mode register **325**. For example, the memory device **120** may detect the PLN signal **315** and may perform the EPF operation based on the mode register **325** being in the low state. Alternatively, the memory device **120** may detect the PLN signal **315** and may perform the FQ operation based on the mode register **325** being in the high state. As described herein, the EPF operation may result in a longer boot-up time during a next powering on of the memory device **120**.

[0038] As indicated above, FIG. **3** is provided as an example. Other examples may differ from what is described with regard to FIG. **3**.

[0039] FIGS. **4A** and **4B** illustrate examples of a two-stage emergency storing operation **400**. The two-stage emergency storing operation **400** may be performed by the memory device **120**, such as by the controller **130** associated with the memory device **120**. A first stage of the two-stage emergency storing operation **400** may include a first data storing operation such as an EPF operation. A second stage of the two-stage emergency storing operation **400** may include a second data storing operation such as an FQ operation. The memory device **120** may receive a power signal. As shown by reference number **405**, the power loss signal (PLS) may not be in a ready state. For example, the memory device **120** may not be experiencing a low battery condition or a disconnected battery condition. As shown by reference number **410**, the PLS may be in a ready state. For example, the memory device **120** may be experiencing the low battery condition or the disconnected battery condition.

[0040] In some cases, as shown in FIG. **4A** and by reference number **415**, an EPF processing port may be enabled. The EPF processing port may be enabled based on the EPF operation being enabled, the PLN signal being asserted, and the memory device **120** supporting the EPF processing port enabled state (Option 1). The memory device **120** may initiate the EPF operation based on the EPF operation being enabled, the PLN signal being asserted, and the EPF processing port being enabled. As shown by reference number **420**, the memory device **120** may complete the EPF operation, and an EPF complete port may be enabled based on the EPF operation being completed. The memory device **120** may perform a controller level reset based on the EPF operation being complete and the EPF complete port being enabled. In some cases, as shown by reference number **425**, an EPF processing port may be disabled. The EPF processing port may be disabled based on the EPF operation being enabled, the PLN signal being asserted, and the memory device **120** supporting the EPF processing port disabled state (Option 2). As shown by reference number **430**, the memory device **120** may complete the EPF operation, and the EPF complete port may be disabled based on the EPF operation being completed. In this case, the memory device **120** may not perform a controller level reset. In some cases, as shown by reference number **435**, FQ processing may be enabled. The FQ processing may be enabled based on the FQ operation being enabled and the PLN signal being asserted (Option 3). As shown by reference number **440**, the memory device **120** may complete the FQ operation. The memory device **120** may perform a controller level reset based on the FQ operation being complete. In some cases, the PLN signal may be de-asserted based on the memory device completing the FQ operation.

[0041] As described in the preceding paragraph, the EPF operation or the FQ operation may be initiated or performed based on one or more conditions. For example, the EPF operation may be performed based on EPF being enabled, such as based on a register bit indicating for the memory device **120** to perform the EPF operation. Alternatively, the FQ operation may be performed based on the FQ operation being enabled, such as based on the register bit indicating for the memory device **120** to perform the FQ operation.

[0042] In some implementations, the memory device **120** may be configured with a second register bit, such as an EPF_FQ bit. When the EPF_FQ bit is in a first state (such as a low state), the memory device **120** may be configured to perform one of the EPF operation or the FQ operation (as described above). For example, the memory device **120** may perform the EPF operation or the

FQ operation in accordance with the mode register **325**. When the EPF_FQ bit is in a second state (such as a high state), the memory device **120** may be configured to perform a storing operation that includes the EPF operation and the FQ operation.

[0043] In some implementations, as shown in FIG. **4B** and by reference number **445**, the PLS may be in a ready state. The memory device **120** may detect a PLN signal and may determine that an EPF_FQ mode is enabled. As shown by reference number **450**, the memory device **120** may initiate EPF processing. For example, the memory device **120** may store data (such as in-flight data) associated with the memory device **120**. As shown by reference number **455**, the memory device **120** may complete the EPF operation. The memory device **120** may complete the EPF operation prior to a first time, such as an EPF vault time (EPFVT). The memory device **120** may complete the EPF operation based on the memory device data (such as the in-flight memory device data) being stored prior to the expiration of the EPFVT.

[0044] As shown by reference number **460**, the memory device **120** may perform (e.g., continue) the FQ operation. For example, the memory device **120** may store metadata, such as metadata associated with data that was stored during the EPF operation. The memory device **120** may perform the FQ operation based on a completion of the EPF operation prior to the first time and based on the memory device **120** having enough power to continue with the FQ operation. As shown by reference number **465**, the memory device **120** may complete the FQ operation (and the EPF operation). The memory device **120** may complete the FQ operation prior to a second time, such as an FQ vault time (FQVT). The memory device **120** may complete the EPF operation and the FQ operation based on memory device data and memory device metadata being stored in the memory device **120** prior to the expiration of the FQVT.

[0045] In some implementations, the memory device **120** may perform the EPF operation and the FQ operation based on detecting a power loss condition. For example, the memory device **120** may receive the PLN signal **315**, initiate the EPF processing **450**, complete the EPF processing **455**, continue the FQ processing **460**, and complete the FQ processing **465** based on receiving the PLN signal **315**. In some implementations, the memory device **120** may perform the EPF operation and the FQ operation based on a standard shut-down of the memory device **120**. For example, the memory device **120** may detect a standard shut-down operation of the memory device **120**, may initiate the EPF processing **450**, complete the EPF processing **455**, continue the FQ processing **460**, and complete the FQ processing **465** based on detecting the standard shut-down operation. In some implementations, the memory device **120** may perform the combined saving operation (EPF and FQ) based on the power loss condition and the standard shut-down condition in parallel. For example, the memory device **120** may perform the EPF processing using energy from the capacitors. Thus, in the event of an emergency power fail, the memory device **120** may save critical data prior to experiencing the power loss. In the case of a standard (e.g., regular) shut-down operation, the memory device **120** may receive the PLN signal **315** from the power management component **305** and/or other circuitry. However, in this case, the power may be maintained for the memory device **120** to perform the FQ processing and the EPF processing prior to the power being cut-off. After the PLA is asserted by the memory device **120**, the voltages that supply power to the memory device **120** may be cut off, thereby completing the shut-down.

[0046] In some implementations, the memory device **120** may perform the EPF operation and the FQ operation based on power received from one or more capacitors, such as the capacitors described in connection with FIG. **6**. In some implementations, the memory device **120** may perform a controller level reset based on completing the EPF operation and the FQ operation. In some implementations, the memory device **120** may assert a PLA signal based on completing the EPF operation and the FQ operation. In some implementations, the memory device **120** may de-assert the PLN signal based on completing the EPF operation and the FQ operation. As described herein, the memory device **120** may perform the EPF operation to save data prior to the memory device **120** experiencing the power loss, and may continue to perform the FQ operation upon a

completion of the EPF operation. This may enable the memory device **120** to store data and metadata prior to the memory device **120** experiencing a power loss, thereby preventing a longer a boot-up time during a next powering on of the memory device.

[0047] As indicated above, FIGS. **4A** and **4B** are provided as examples. Other examples may differ from what is described with regard to FIGS. **4A** and **4B**.

[0048] FIG. **5** is a diagrammatic view of an example memory device **500**. The memory device **500** may include a memory array **502** that includes multiple memory cells **504**. A memory cell **504** is programmable or configurable into a data state of multiple data states (e.g., two or more data states). For example, a memory cell **504** may be set to a particular data state at a particular time, and the memory cell **504** may be set to another data state at another time. A data state may correspond to a value stored by the memory cell **504**. The value may be a binary value, such as a binary 0 or a binary 1, or may be a fractional value, such as 0.5, 1.5, or the like. A memory cell **504** may include a capacitor to store a charge representative of the data state. For example, a charged and an uncharged capacitor may represent a first data state and a second data state, respectively. As another example, a first level of charge (e.g., fully charged) may represent a first data state, a second level of charge (e.g., fully discharged) may represent a second data state, a third level of charge (e.g., partially charged) may represent a third data state, and so on.

[0049] Operations such as reading and writing (i.e., cycling) may be performed on memory cells **504** by activating or selecting the appropriate access line **506** (shown as access lines AL **1** through AL **M**) and digit line **508** (shown as digit lines DL **1** through DL **N**). An access line **506** may also be referred to as a “row line” or a “word line,” and a digit line **508** may also be referred to a “column line” or a “bit line.” Activating or selecting an access line **506** or a digit line **508** may include applying a voltage to the respective line. An access line **506** and/or a digit line **508** may comprise, consist of, or consist essentially of a conductive material, such as a metal (e.g., copper, aluminum, gold, titanium, or tungsten) and/or a metal alloy, among other examples. In FIG. **5**, each row of memory cells **504** is connected to a single access line **506**, and each column of memory cells **504** is connected to a single digit line **508**. By activating one access line **506** and one digit line **508** (e.g., applying a voltage to the access line **506** and digit line **508**), a single memory cell **504** may be accessed at (e.g., is accessible via) the intersection of the access line **506** and the digit line **508**. The intersection of the access line **506** and the digit line **508** may be called an “address” of a memory cell **504**.

[0050] In some implementations, the logic storing device of a memory cell **504**, such as a capacitor, may be electrically isolated from a corresponding digit line **508** by a selection component, such as a transistor. The access line **506** may be connected to and may control the selection component. For example, the selection component may be a transistor, and the access line **506** may be connected to the gate of the transistor. Activating the access line **506** results in an electrical connection or closed circuit between the capacitor of a memory cell **504** and a corresponding digit line **508**. The digit line **508** may then be accessed (e.g., is accessible) to either read from or write to the memory cell **504**.

[0051] A row decoder **510** and a column decoder **512** may control access to memory cells **504**. For example, the row decoder **510** may receive a row address from a memory controller **514** and may activate the appropriate access line **506** based on the received row address. Similarly, the column decoder **512** may receive a column address from the memory controller **514** and may activate the appropriate digit line **508** based on the column address.

[0052] Upon accessing a memory cell **504**, the memory cell **504** may be read (e.g., sensed) by a sense component **516** to determine the stored data state of the memory cell **504**. For example, after accessing the memory cell **504**, the capacitor of the memory cell **504** may discharge onto its corresponding digit line **508**. Discharging the capacitor may be based on biasing, or applying a voltage, to the capacitor. The discharging may induce a change in the voltage of the digit line **508**, which the sense component **516** may compare to a reference voltage (not shown) to determine the

stored data state of the memory cell **504**. For example, if the digit line **508** has a higher voltage than the reference voltage, then the sense component **516** may determine that the stored data state of the memory cell **504** corresponds to a first value, such as a binary 1. Conversely, if the digit line **508** has a lower voltage than the reference voltage, then the sense component **516** may determine that the stored data state of the memory cell **504** corresponds to a second value, such as a binary 0. The detected data state of the memory cell **504** may then be output (e.g., via the column decoder **512**) to an output component **518** (e.g., a data buffer). A memory cell **504** may be written (e.g., set) by activating the appropriate access line **506** and digit line **508**. The column decoder **512** may receive data, such as input from input component **520**, to be written to one or more memory cells **504**. A memory cell **504** may be written by applying a voltage across the capacitor of the memory cell **504**.

[0053] The memory controller **514** may control the operation (e.g., read, write, re-write, refresh, and/or recovery) of the memory cells **504** via the row decoder **510**, the column decoder **512**, and/or the sense component **516**. The memory controller **514** may generate row address signals and column address signals to activate the desired access line **506** and digit line **508**. The memory controller **514** may also generate and control various voltages used during the operation of the memory array **502**.

[0054] As indicated above, FIG. 5 is provided as an example. Other examples may differ from what is described with respect to FIG. 5.

[0055] FIG. 6 is a flowchart of an example method **600** associated with a two-stage emergency data storing operation. In some implementations, a memory device (e.g., the memory device **120**) may perform or may be configured to perform the method **600**. In some implementations, another device or a group of devices separate from or including the memory device (e.g., the system **100**) may perform or may be configured to perform the method **600**. Additionally, or alternatively, one or more components of the memory device (e.g., the controller **130**, the memory management component **225**, emergency power fail component **230**, forced quiescence component **235**, and/or emergency storing component **240**) may perform or may be configured to perform the method **600**. Thus, means for performing the method **600** may include the memory device and/or one or more components of the memory device. Additionally, or alternatively, a non-transitory computer-readable medium may store one or more instructions that, when executed by the memory device (e.g., the controller **130** of the memory device **120**), cause the memory device to perform the method **600**.

[0056] As shown in FIG. 6, the method **600** may include reading a mode register bit that indicates to perform a data storing operation that includes an emergency power fail operation and a forced quiescence operation (block **610**). As further shown in FIG. 6, the method **600** may include initiating and completing the emergency power fail operation (block **620**). As further shown in FIG. 6, the method **600** may include initiating the forced quiescence operation based on completing the emergency power fail operation (block **630**). As further shown in FIG. 6, the method **600** may include asserting a power loss acknowledgement based on completing the forced quiescence operation (block **640**).

[0057] The method **600** may include additional aspects, such as any single aspect or any combination of aspects described below and/or described in connection with one or more other methods or operations described elsewhere herein.

[0058] In a first aspect, a first state of the mode register bit indicates for the memory device to perform the data storing operation that includes the emergency power fail operation and the forced quiescence operation, and a second state of the mode register bit indicates for the memory device to perform a data storing operation that includes only one of the emergency power fail operation or the forced quiescence operation.

[0059] In a second aspect, alone or in combination with the first aspect, the method **600** includes detecting a power loss condition or a standard shut-down condition of the memory device, wherein

reading the mode register bit comprises reading the mode register bit based on detecting the power loss condition or the standard shut-down condition of the memory device.

[0060] In a third aspect, alone or in combination with one or more of the first and second aspects, the power loss condition is associated with a low battery condition or a disconnected battery condition.

[0061] In a fourth aspect, alone or in combination with one or more of the first through third aspects, initiating and completing the emergency power fail operation, and initiating the forced quiescence operation, comprises initiating and completing the emergency power fail operation, and initiating the forced quiescence operation, using power received from one or more capacitors associated with the memory device.

[0062] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, asserting the power loss acknowledgement comprises transmitting a power loss acknowledgement signal to a power management component.

[0063] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, transmitting the power loss acknowledgement signal comprises transmitting the power loss acknowledgement signal to the power management component based on completing the emergency power fail operation and the forced quiescence operation.

[0064] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the method **600** includes performing a controller level reset or de-asserting a power loss notification signal based on completing the emergency power fail operation and the forced quiescence operation.

[0065] Although FIG. **6** shows example blocks of a method **600**, in some implementations, the method **600** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of the method **600** may be performed in parallel. The method **600** is an example of one method that may be performed by one or more devices described herein. These one or more devices may perform or may be configured to perform one or more other methods based on operations described herein.

[0066] In some implementations, a memory device includes one or more components configured to: detect a power loss notification signal that indicates a power loss condition of the memory device; read a mode register bit of the memory device that indicates to perform a data storing operation that includes a first data storing stage and a second data storing stage, wherein the first data storing stage includes storing data associated with the memory device prior to the memory device experiencing a power loss, and the second data storing stage includes storing data and metadata associated with the memory device prior to the memory device experiencing the power loss; initiate the data storing operation; and selectively acknowledge the power loss condition based on completing the first data storing stage or the second data storing stage.

[0067] In some implementations, a system includes a power management component configured to: detect a power loss condition associated with a memory device; and transmit a power loss notification signal to the memory device that indicates the power loss condition; and the memory device, wherein the memory device is configured to: receive the power loss notification signal; read a mode register bit of the memory device that indicates to perform a data storing operation that includes a first data storing stage and a second data storing stage, wherein the first data storing stage includes storing data associated with the memory device prior to the memory device experiencing a power loss, and the second data storing stage includes storing data and metadata associated with the memory device prior to the memory device experiencing the power loss; initiate the data storing operation; and selectively acknowledge the power loss condition based on completing the first data storing stage or the second data storing stage.

[0068] In some implementations, a method performed by a memory device includes detecting a power loss condition of the memory device; reading a mode register bit that indicates to perform a data storing operation that includes an emergency power fail operation and a forced quiescence

operation; initiating and completing the emergency power fail operation; initiating the forced quiescence operation based on completing the emergency power fail operation; and asserting a power loss acknowledgement based on completing the forced quiescence operation.

[0069] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations described herein.

[0070] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of implementations described herein. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. For example, the disclosure includes each dependent claim in a claim set in combination with every other individual claim in that claim set and every combination of multiple claims in that claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

[0071] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Where only one item is intended, the phrase “only one,” “single,” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. As used herein, the term “multiple” can be replaced with “a plurality of” and vice versa. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A memory device, comprising: one or more components configured to: read a mode register bit of the memory device, wherein a first state of the mode register bit indicates to perform a first data storing operation that includes a first stage and a second stage, wherein a second state of the mode register bit indicates to perform a second data storing operation that includes only one of the first stage or the second stage, and wherein the first stage includes storing data prior to the memory device experiencing a power loss, and the second stage includes storing data and metadata prior to the memory device experiencing the power loss; and initiate the first data storing operation or the second data storing operation based on the mode register bit.
2. The memory device of claim 1, wherein the first stage includes an emergency power fail operation and the second stage includes a forced quiescence operation.
3. The memory device of claim 1, wherein the one or more components are further configured to: detect a power loss notification signal that indicates a power loss condition of the memory device, wherein reading the mode register bit is based on detecting the power loss notification signal.
4. The memory device of claim 1, wherein, when the mode register bit is in the second state, the one or more components are further configured to: read an additional mode register bit, wherein a first state of the additional mode register bit indicates that the first stage is enabled, and wherein a second state of the additional mode register bit indicates that the second stage is enabled.

5. The memory device of claim 4, wherein the one or more components are further configured to: initiate the second data storing operation including the first stage or the second stage based on the mode register bit and the additional mode register bit.
6. The memory device of claim 1, wherein, when the mode register bit is in the first state, the one or more components are further configured to: perform the first stage of the first data storing operation; and initiate the second stage of the first data storing operation based on completing the first stage prior to a first time and based on having enough power to continue with the second stage.
7. The memory device of claim 1, wherein the one or more components are configured to perform the first data storing operation or the second data storing operation using power received from one or more capacitors associated with the memory device.
8. The memory device of claim 1, wherein the one or more components are configured to perform the first data storing operation or the second data storing operation using one or more voltages that supply power to the memory device.
9. A system, comprising: a power component configured to transmit a signal, to a memory device, that indicates a power loss condition; and the memory device, wherein the memory device is configured to: read, based on the signal, a mode register bit of the memory device, wherein a first state of the mode register bit indicates to perform a first data storing operation that includes a first stage and a second stage, wherein a second state of the mode register bit indicates to perform a second data storing operation that includes only one of the first stage or the second stage, and wherein the first stage includes storing data prior to the memory device experiencing a power loss, and the second stage includes storing data and metadata prior to the memory device experiencing the power loss; and initiate the first data storing operation or the second data storing operation based on the mode register bit and the signal.
10. The system of claim 9, wherein the first stage includes an emergency power fail operation and the second stage includes a forced quiescence operation.
11. The system of claim 9, wherein, when the mode register bit is in the second state, the memory device is further configured to: read an additional mode register bit, wherein a first state of the additional mode register bit indicates that the first stage is enabled, and wherein a second state of the additional mode register bit indicates that the second stage is enabled.
12. The system of claim 11, wherein the memory device is further configured to: initiate the second data storing operation including the first stage or the second stage based on the mode register bit and the additional mode register bit.
13. The system of claim 9, wherein, when the mode register bit is in the first state, the memory device is further configured to: perform the first stage of the first data storing operation; and initiate the second stage of the first data storing operation based on completing the first stage prior to a first time.
14. A method performed by a memory device, comprising: reading a mode register bit, wherein a first state of the mode register bit indicates to perform a first data storing operation that includes a first stage and a second stage, wherein a second state of the mode register bit indicates to perform a second data storing operation that includes only one of the first stage or the second stage, and wherein the first stage includes storing data prior to the memory device experiencing a power loss, and the second stage includes storing data and metadata prior to the memory device experiencing the power loss; and initiating the first data storing operation or the second data storing operation based on the mode register bit.
15. The method of claim 14, wherein the first stage includes an emergency power fail operation and the second stage includes a forced quiescence operation.
16. The method of claim 14, further comprising detecting a power loss condition or a standard shut-down condition of the memory device, wherein reading the mode register bit comprises reading the mode register bit based on detecting the power loss condition or the standard shut-down condition of the memory device.

17. The method of claim 14, further comprising detecting a power loss notification signal that indicates a power loss condition of the memory device, wherein reading the mode register bit is based on detecting the power loss notification signal.

18. The method of claim 14, wherein the mode register bit is in the second state, the method further comprising: reading an additional mode register bit, wherein a first state of the additional mode register bit indicates that the first stage is enabled, and wherein a second state of the additional mode register bit indicates that the second stage is enabled.

19. The method of claim 14, wherein the first data storing operation or the second data storing operation is performed using power received from one or more capacitors associated with the memory device.

20. The method of claim 14, wherein the first data storing operation or the second data storing operation is performed using one or more voltages that supply power to the memory device.
